JAYWARDHAN YADAV

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Summary

AI/ML Engineer (M.Sc. Data Science, CGPA 9.60) building production-grade **agentic AI and RAG systems**. Shipped a **multi-agent LangGraph planner** with human-in-the-loop and MCP (96.0% guardrail accuracy, ~$0.018 per trip) and a **hybrid-search legal RAG platform** that lifted Hit Rate@5 from 48.2% to 86.8% on a 75-query benchmark. Strong in **FastAPI, PostgreSQL, Docker** and MLOps (**MLflow, DVC**), taking systems from data and models to deployed APIs.

Technical Skills

Generative AI & Agentic AI: Agentic AI, Multi-Agent Orchestration (LangGraph, LangChain), Tool / Function Calling, Model Context Protocol (MCP), Human-in-the-Loop Workflows, RAG, Hybrid Search (BM25 + Dense), Cross-Encoder Reranking, Embeddings, Weaviate, Prompt Engineering, LLM APIs (OpenAI, Groq), Hugging Face, RAG Evaluation (Ragas), LLM Tracing (LangSmith)

ML & Data Science: Python, PyTorch, Scikit-learn, NumPy, Pandas, EDA, Feature Engineering, Statistical Analysis, Regression, Random Forest, XGBoost, LightGBM, Optuna

MLOps, Cloud & DevOps: AWS (S3), MLflow, DVC, Evidently AI (Data Drift Monitoring), Docker, CI/CD, Git/GitHub, Alembic Migrations

Backend & Databases: FastAPI (REST), Pydantic, JWT Authentication, Async Programming, Server-Sent Events (SSE), Streamlit, SQL, PostgreSQL, MSSQL / SQL Server, SQLAlchemy, Redis

Projects

PlanMyTrip AI – Multi-Agent Travel Planning System with Human-in-the-Loop github.com/JaywardhanYadav/PlanMyTrip-AI
Technologies: Python, LangGraph, FastAPI, OpenAI (GPT-4o-mini), LangSmith, PostgreSQL, Redis, MCP, Pydantic V2, Docker, SSE, JWT

- Architected a two-stage **multi-agent travel planner in LangGraph** (parallel fan-out agents, then location-grounded hotel matching) with **Human-in-the-Loop checkpoints**; decoupled live flight, hotel, attraction, and weather data behind **3 standalone MCP servers**.
- Built an input guardrail and **evaluation suite** over **10 trip scenarios and 25 adversarial cases**: **96.0%** guardrail accuracy, **96.8%** budget compliance, **95.5%** itinerary quality, **11.4s p50 / 18.2s p95** latency.
- Engineered reliable agent state with **Pydantic V2**, an **idempotent keyed-merge reducer**, and *Decimal* budgets (zero floating-point drift); kept LLM cost to **~$0.018 (INR 1.50) per trip** across 5 GPT-4o-mini agents, traced in **LangSmith**.
- Built an async **FastAPI** backend with **JWT** auth, **PostgreSQL** checkpoints, and per-user thread ownership; streamed progress over **SSE**; containerized with **Docker** and covered by an end-to-end smoke test.

LegalDrishti AI – Two-Stage RAG Legal Research Platform (Indian Law) github.com/JaywardhanYadav/LegalDrishti-AI
Technologies: Python, FastAPI, PostgreSQL, SQLAlchemy (AsyncIO), Alembic, Weaviate, OpenAI Embeddings, FlashRank, Ragas, Redis, Docker

- Architected a **Two-Stage RAG pipeline** with a custom **Statute Router** (regex intent + weighted n-gram index, IPC/CrPC to BNS/BNSS/BSA mapping) that pre-filters **28 Indian Bare Acts**; **Weaviate hybrid search** (dense + BM25, alpha=0.5) fetches 20 candidates, then a **FlashRank cross-encoder** reranks to the top 5 for answers with Act, Section and page-level citations.
- Benchmarked on a **75-query golden set**: **Hit Rate@5 48.2% to 86.8%** and **MRR@5 0.384 to 0.762 (+98%)** over dense-only search at ~68 ms retrieval latency; **Ragas** faithfulness 88.4%, context precision 84.2%, answer relevancy 85.6%.
- Built an async **FastAPI + PostgreSQL (SQLAlchemy AsyncIO)** backend with JWT auth, metered Deep Search credits, and dual-stream grounding over statutes and the user's **document vault**; ephemeral chat attachments auto-purge on chat deletion and a master purge wipes files and embeddings; containerized with Docker.

Health Insurance Premium Prediction & End-to-End MLOps Pipeline github.com/JaywardhanYadav/Insurance-premium-prediction-mlops
Technologies: Python, XGBoost, LightGBM, Optuna, MLflow, DVC, FastAPI, Pydantic, Streamlit, MSSQL, SQLAlchemy, Evidently AI, Docker

- Built a config-driven (YAML) **end-to-end ML pipeline**: clinical feature engineering (blood-pressure parsing, outlier clipping, interaction features such as **risk_score, smoker_obese, hypertension**) and an **Optuna-tuned XGBoost + LightGBM ensemble** trained on **20K records** with a log-transformed premium (train/val/test split): **test R2 0.854, RMSE 2,891, MAE 2,201**; runs and models tracked in **MLflow**.
- Served predictions through a **FastAPI** inference API with **Pydantic** request validation and a **Streamlit** dashboard; every prediction and its latency is logged to **MSSQL** through a **SQLAlchemy** data-access layer.
- Added an **Evidently AI drift monitor** that compares production prediction logs against a reference baseline as a containerized job; versioned data with **DVC**, packaged the API, UI, database and monitor as a multi-container **Docker** stack, and wrote an AWS ECS Fargate deployment guide.

Education

Master in Data Science – CGPA – 9.60 *2024 – 2026*
Dr. D. Y. Patil College, Pimpri, Pune

Bachelor in Computer Science – CGPA – 8.40 *2021 – 2024*
Nowrosjee Wadia College, Pune

Awards

- Secured **1st Rank** at the college level for our **research paper** (2025), which was selected for the **university research publication**.
- Secured **2nd Rank** in the college **hackathon (programming competition)** in 2026.
- Awarded the **Silver Medal** in **M.Sc. Data Science** at Dr. D. Y. Patil College (2026).

Certifications

- Generative AI & Agentic AI – **Udemy** | Agentic AI Certified Foundations Associate – **Oracle** | Full Stack Data Science – **Udemy**
- RAG Architecture – **DeepLearning.AI**
- Complete Generative AI Course with LangChain and Hugging Face – **Krish Naik Academy**
- Solved **100+ DSA and Data Science problems** on **LeetCode** and **HackerRank**